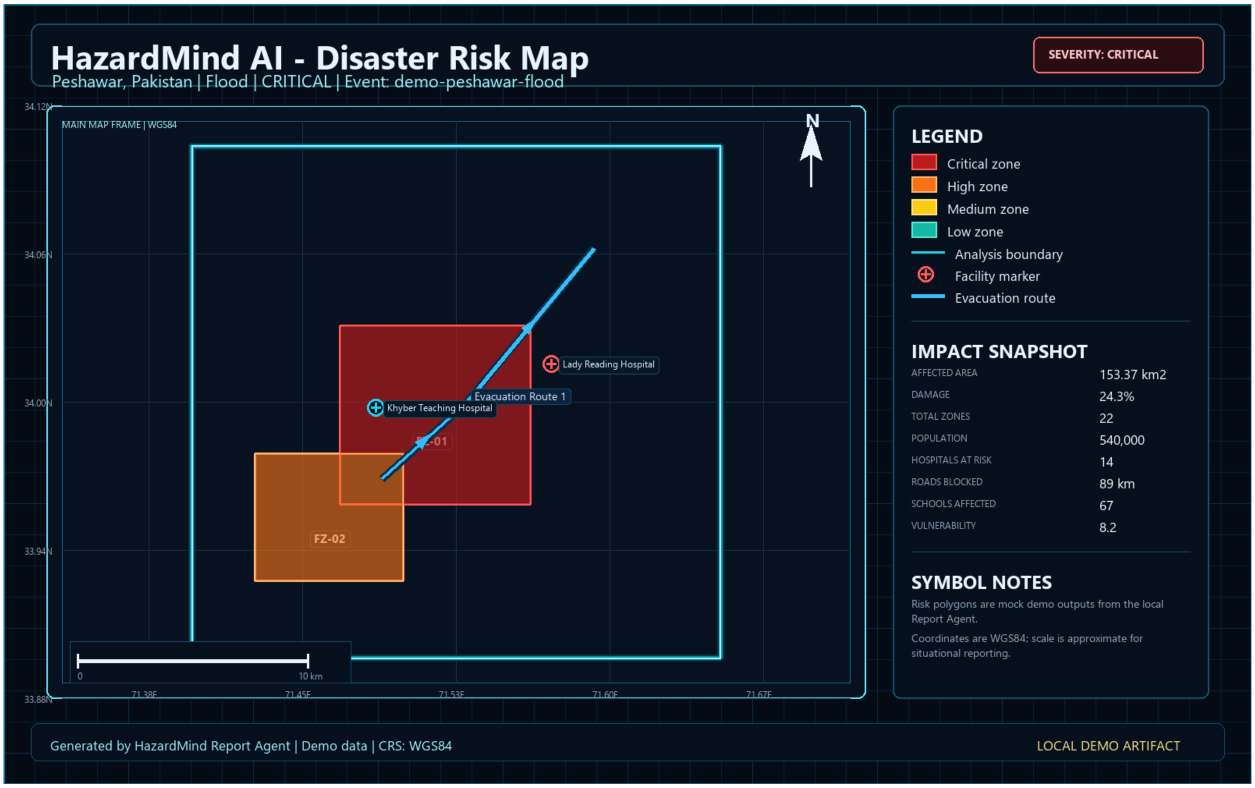
HazardMind AI Executive Disaster Report

Event ID	demo-peshawar-flood
Location	Peshawar, Pakistan
Hazard Type	Flood
Overall Severity	CRITICAL
Satellite Type	sentinel-1
Satellite Reason	cloud_cover_above_30_percent_sar_selected

Operational Statistics

Affected Area	153.37 km2
Damage Percent	24.3%
Total Zones	22
Population Affected	540,000
Hospitals at Risk	14
Roads Blocked	89 km
Schools Affected	67
Vulnerability Score	8.2

Generated Risk Map



Executive Summary

Peshawar faces a CRITICAL flood emergency with 153.37 km2 inundated and 540,000 people affected, confirmed via Sentinel-1 SAR at 0.91 confidence. A single available evacuation route severely constrains egress from eastern sectors, and 89 km of blocked roads sever primary logistics. Deep-water Zone FZ-01 threatens Lady Reading Hospital, with 14 hospitals and 67 schools impacted against a high vulnerability score of 8.2. Medium earthquake risk warrants monitoring as a cascading hazard. All figures are decision-support indicators pending field validation, and missing report/map links require correction before distribution.

Intelligence Assessment

Criticality	critical
Overall Confidence	91%
Escalation Required	True
Rationale	Critical flood emergency in Peshawar with 153 km² inundated, 540,000 people exposed, 14 hospitals at risk
Key Decisions	Authorize immediate evacuation of flood zones FZ-01 and FZ-02 and secure access to Lady Reading Hosp
Human Review Required	True

Map Narrative

The static risk map identifies a critical flood emergency in Peshawar, confirming 153.37 km² of inundation across 22 zones. With a high vulnerability score of 8.2 and over 540,000 people affected, the map highlights a systemic failure in mobility due to 89 km of blocked roads and a single remaining evacuation route, creating an urgent operational bottleneck for rescue and egress.

- Critical flood inundation across 22 distinct zones
- 89 km of road network blocked, severing primary logistics
- Single viable evacuation route constraining egress from eastern sectors
- High spatial correlation between flood zones and critical healthcare facilities

Priority Timeline

Next 6 Hours	Evacuate high-risk flood zones FZ-01 and FZ-02 immediately; Secure access routes to Lady Reading Hosp
Next 24 Hours	Clear 89km of blocked roads to restore logistics; Establish temporary shelters for 540,000 affected residen
Next 72 Hours	Begin structural damage assessments for 67 affected schools; Restore primary utility services in inundated
Resources	High-capacity water rescue boats and amphibious vehicles; Emergency medical supplies for critical care fa
Coordination	Synchronize evacuation flow on the single available route; Align health department resources with at-risk h

Anomalies and Warnings

- MEDIUM: Report PDF and Map URLs are empty. Handling: Re-run report generation module or inject manual links before distribution.
- LOW: Zone feature count (2) is significantly lower than total zones analyzed (22). Handling: Verify zoning extraction logic and re-run feature extraction to ensure comprehensive coverage.
- GeoJSON artifact URL is not available in local demo mode.
- Map URL is generated later by the local artifact step.
- Report PDF and map URLs are empty, requiring regeneration before distribution.
- Zone feature count is 0 across 22 analyzed zones, reducing spatial validation confidence.
- Evacuation route count shows discrepancy between route object (0) and narrative (1).
- Medium earthquake risk requires ongoing monitoring for cascading effects.

Quality Check

Status	not_ready
Event Id Present	True
Satellite Data Present	True
Hazard Data Present	True
Impact Data Present	True
Map Artifacts Present	True
Recommendations Present	True
Confidence Above Threshold	True

Warnings	GeoJSON artifact URL is not available in local demo mode.; Map URL is generated later by the local artifac
Blocking Issues	Missing report PDF and map URLs block official distribution and stakeholder communication.; Contradictor

Band-Ready Final Message

HazardMind Report Agent complete for demo-peshawar-flood (Peshawar, Pakistan, Flood). Criticality: critical; confidence: 91%. Summary: Peshawar faces a CRITICAL flood emergency with 153.37 km2 inundated and 540,000 people affected, confirmed via Sentinel-1 SAR at 0.91 confidence. A single available evacuation route severely constrains egress from eastern sectors, and 89 km of blocked roads sever primary logistics. Deep-water Zone FZ-01 threatens Lady Reading Hospital, with 14 hospitals and 67 schools impacted against a high vulnerability score of 8.2. Medium earthquake risk warrants monitoring as a cascading hazard. All figures are decision-support indicators pending field validation, and missing report/map links require correction before distribution.

Detailed Incident Analysis

Peshawar, Pakistan is experiencing a CRITICAL flood scenario with 153.37 km2 affected and 540,000 people exposed. Flood risk is classified as CRITICAL with hospitals, roads, and schools requiring immediate operational attention.

Technical Analysis

Sentinel-1 SAR analysis reveals 153.37 km² of floodwater across 22 zones, with 24.3% damage and a mean VV/VH ratio of 0.24 indicating extensive inundation. Critical deep water zones (FZ-01: 12.4 km²) combine with high social impact: 540,000 affected, 14 hospitals at risk (Lady Reading Hospital HIGH risk), 89 km of roads blocked, and 67 schools affected. Vulnerability score 8.2 and a single evacuation route amplify response challenges. Potential cascading hazard from medium earthquake risk further complicates the crisis.

Recommendations

- Prioritize evacuation in critical flood zones.
- Deploy emergency medical support near hospitals at risk.
- Clear blocked road corridors for rescue access.
- Open temporary shelters for displaced families.
- Monitor flood expansion using updated satellite imagery.
- Evacuate high-risk flood zones FZ-01 and FZ-02 immediately
- Secure access routes to Lady Reading Hospital
- Deploy emergency rescue teams to critical deep-water zones
- High-capacity water rescue boats and amphibious vehicles
- Emergency medical supplies for critical care facilities

Response Priorities

- Activate evacuation support for FZ-01 and FZ-02.
- Protect hospital access and stage medical surge teams near high-risk facilities.

- Clear blocked road corridors needed for rescue and logistics movement.

Assumptions

- Satellite, hazard, and impact values are local demo data until upstream agents publish live outputs.
- Risk zone geometries are simplified mock polygons for dashboard and report demonstration.
- Facility and route coordinates are approximate and intended for prototype visualization.

Limitations

- No live Band messages are consumed in this local demo run.
- No field validation, hydrologic model, or live basemap tiles are used for artifact generation.
- Impact estimates should be treated as decision-support indicators, not official counts.

Model Source Note

Detailed report generated using Featherless Kimi K2.6. Executive summary generated using AI/ML API. Intelligence sources: criticality: featherless:deepseek-v4-pro, anomaly_check: featherless:qwen-3.6-35b-a3b, map_narrative: featherless:gemma-4-31b-it, priority_recommendations: featherless:gemma-4-31b-it, decision_brief: aiml:opus-4.8, quality_check: hybrid:deterministic+featherless:qwen-3.6-35b-a3b, band_ready_message: template+intelligence

Detailed Report	featherless:composite
Executive Summary	aiml:opus-4.8
Fallback Used	False
Featherless Model	moonshotai/Kimi-K2.6
Criticality	featherless:deepseek-v4-pro
Map Narrative	featherless:gemma-4-31b-it
Priority Timeline	featherless:gemma-4-31b-it
Quality Check	hybrid:deterministic+featherless:qwen-3.6-35b-a3b

Agent Log / Trace

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:00Z	Report Agent received event context
hazardmind-report	complete	2026-06-13T18:03:20Z	Featherless composite generated detailed disaster report
hazardmind-report	complete	2026-06-13T18:03:40Z	AI/ML generated executive summary
hazardmind-report	complete	2026-06-13T18:03:45Z	Criticality assessed
hazardmind-report	complete	2026-06-13T18:03:50Z	Anomalies checked

Agent	Status	Timestamp	Message
hazardmind-report	complete	2026-06-13T18:03:55Z	Map narrative generated
hazardmind-report	complete	2026-06-13T18:04:00Z	Priority timeline generated
hazardmind-report	complete	2026-06-13T18:04:05Z	Decision brief generated with Opus
hazardmind-report	complete	2026-06-13T18:04:10Z	Quality check completed
hazardmind-report	complete	2026-06-13T18:04:15Z	Band-ready final message prepared
hazardmind-report	complete	2026-06-13T18:04:00Z	Static cartography map generated locally